

CHAPTER 6

SCREENSHOTS

6.1 Introduction

This chapter presents the visual documentation of the MedRAG AI-Powered Medical Report Analyzer application through a comprehensive collection of screenshots captured from the live system. Each screenshot serves as empirical evidence of the system's functionality, user interface design, and operational behavior across all major modules. The screenshots demonstrate how the theoretical architecture described in previous chapters manifests into a working, interactive healthcare assistant application.

6.2 Medical Document Upload Module

The Upload Module is the entry point of the MedRAG pipeline, accepting medical documents in multiple formats and triggering the full backend processing workflow automatically.

Supported formats: PDF, PNG, JPG, JPEG, TIFF, WEBP — up to 5 MB.

Processing pipeline triggered on upload:

- PyMuPDF handles native PDF text extraction
- Tesseract OCR processes scanned documents and images
- OpenCV preprocessing applies grayscale, thresholding, and noise reduction
- Auto fallback to OCR if PDF returns insufficient text

Key UI elements visible in the screenshot:

- Drag-and-drop upload zone with format guidance
- Live progress bar showing pipeline stages — *Uploading* → *Extracting* → *OCR* → *Chunking* → *Indexing*
- Real-time extracted text preview for user verification
- Error states for unsupported formats or oversized files

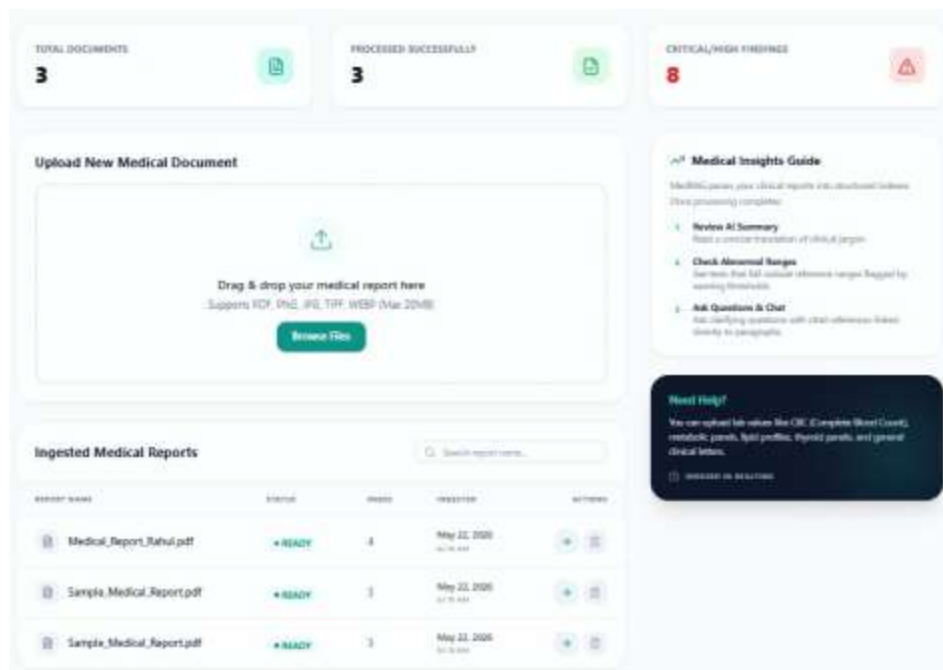


Figure 6.1 Medical Document Upload Module

6.3 Summary Overview

Once a report is indexed, the Summary Overview panel delivers an AI-generated narrative of the report's findings, written in plain language that any patient can understand without medical background.

What the summary includes:

- Key findings interpreted in simple terms
- Co-occurring abnormal patterns identified together (e.g., high glucose + high HbA1c flagged jointly)
- Risk commentary for critical value combinations
- Medical disclaimer advising physician consultation

Key UI elements visible in the screenshot:

- Streamed AI summary rendered in a readable card
- Regenerate button to refresh the output
- Risk flag banner displayed below for critical findings

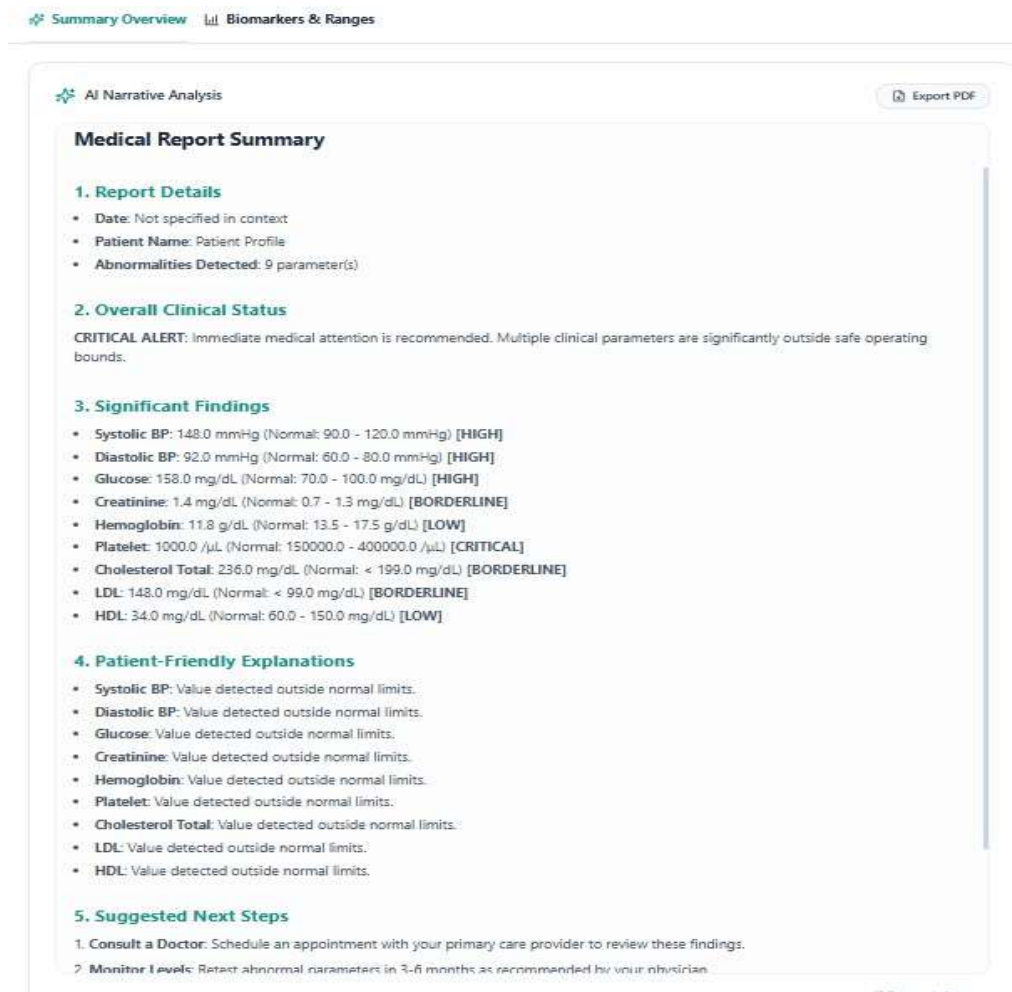


Figure 6.2 Summary Overview

6.4 Export Report PDF

The Export feature allows users to download a professionally formatted AI analysis document suitable for printing or sharing at a clinical consultation.

Exported PDF contains:

- Report metadata — filename, upload date, page count
- Full AI-generated narrative summary
- Color-coded biomarker table with values, normal ranges, and status
- Risk flags for critical findings
- Medical disclaimer footer on every page

Key UI elements visible in the screenshot:

- Export button in the top action bar
- Download confirmation toast notification
- Generated PDF with branded header and color-coded biomarker table

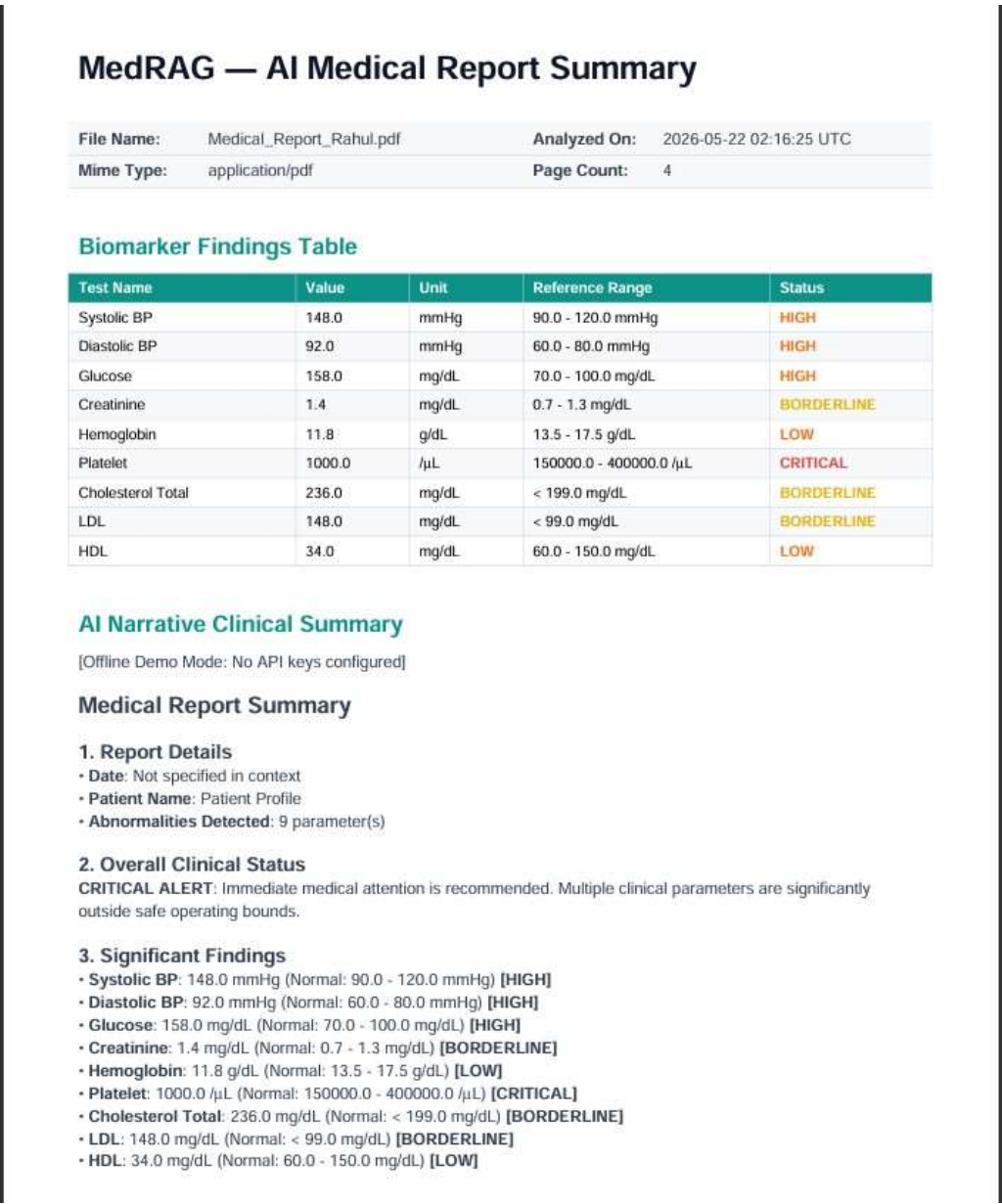


Figure 6.3 Export PDF

6.5 Biomarkers and Reference Ranges

The Biomarkers panel converts raw lab values into a color-coded health dashboard, allowing users to instantly see which results are normal and which require attention.

Status classification system:

Status	Color	Meaning
Normal	 Green	Within clinical reference range
Borderline	 Amber	Slightly outside safe threshold
High / Low	 Orange	Clearly outside normal range
Critical	 Red	Requires immediate medical attention

Biomarkers evaluated:

- Glucose, HbA1c — diabetes markers
- LDL, HDL, Cholesterol, Triglycerides — lipid panel
- WBC, RBC, Hemoglobin, Platelets — CBC
- Creatinine, BUN — kidney function
- TSH, T3, T4 — thyroid panel

Key UI elements visible in the screenshot:

- Per-row color-coded status badge per biomarker
- Normal range displayed beside each detected value
- Bar chart showing each value as a proportion of its normal boundary
- Bars extending past the threshold marker for abnormal values

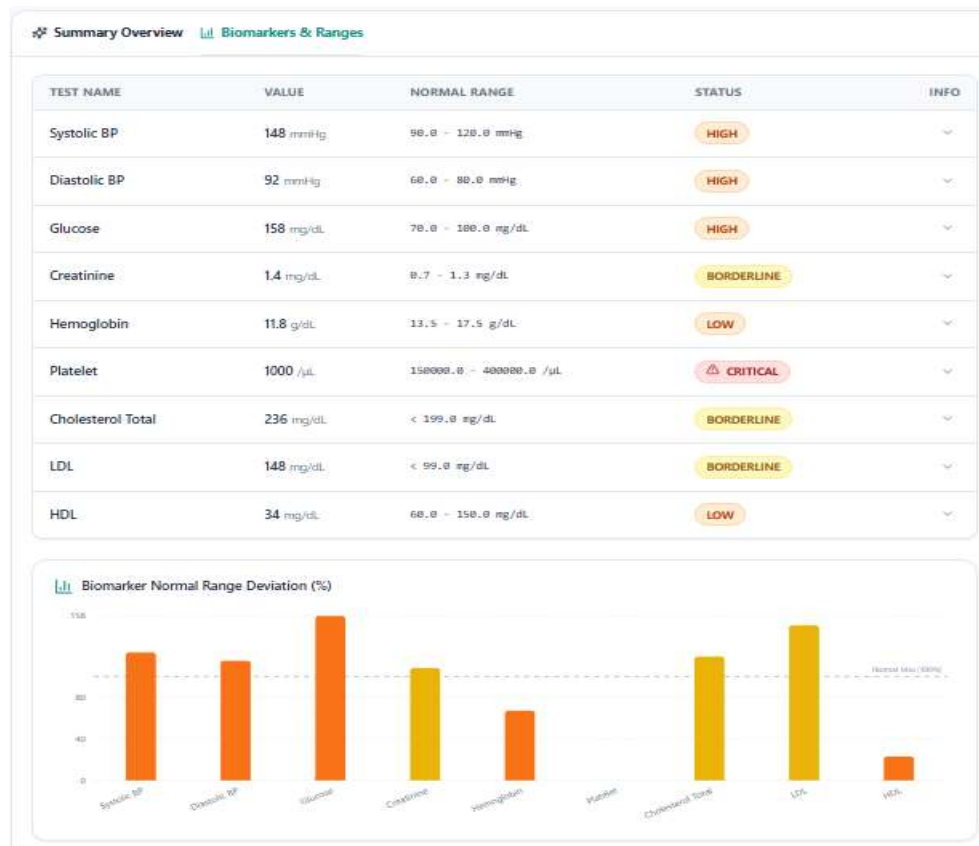


Figure 6.4 Biomarkers and Reference Ranges

6.6 RAG Assistant

The RAG Assistant is the conversational core of MedRAG, allowing users to ask natural language questions about their uploaded report and receive accurate, source-cited answers in real time.

How it works:

- User question is converted to a semantic embedding
- ChromaDB retrieves the top 5 most relevant report chunks
- Chunks + conversation history + medical system prompt are passed to GPT-4o
- Response streams token-by-token via Server-Sent Events
- Source chunk citations are appended below every AI answer

Conversational memory:

- Retains last 10 turns via ConversationBufferWindowMemory

- Supports natural follow-ups without repeating context — *"What is my glucose?"* → *"Is that dangerous?"* → *"What should I eat?"* — all resolved correctly in sequence

Key UI elements visible in the screenshot:

- Streaming chat bubbles with user and assistant messages
- Collapsible source citation chips below each AI response
- Suggested starter questions for first-time users
- Voice input microphone for hands-free querying
- Left-side chat history panel listing all past sessions by report name

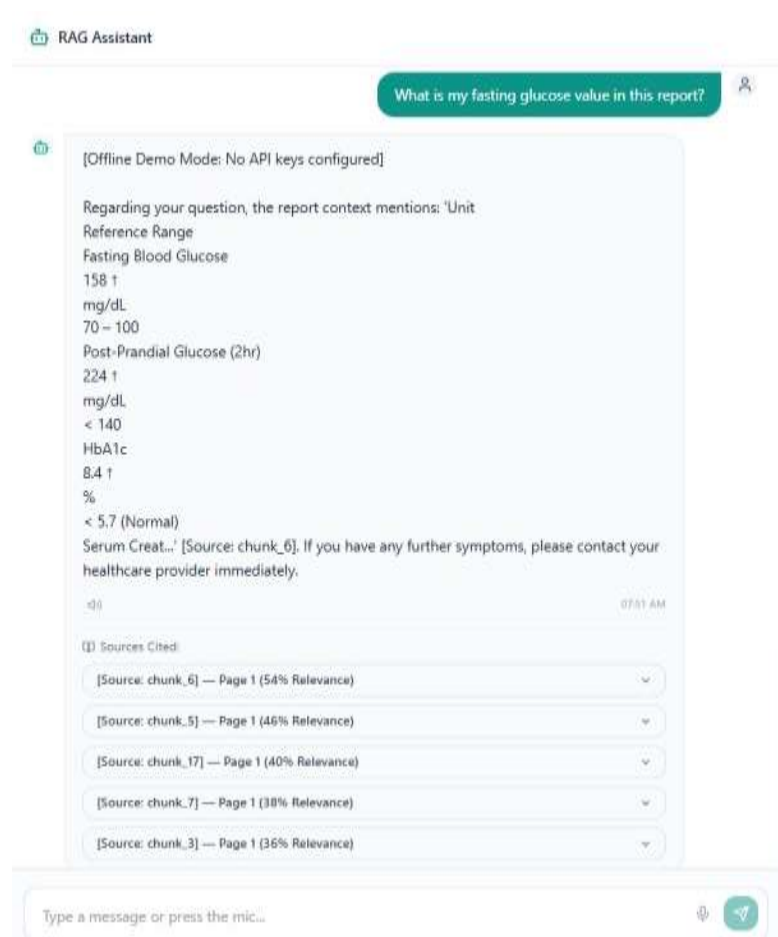


Figure 6.5 RAG Assistant

6.7 Complete System Workflow Pipeline

The MedRAG system follows a fully automated, seven-stage pipeline that transforms a raw uploaded medical document into an intelligent, conversational AI response with zero manual intervention between stages.

Stage 1 — Document Submission User uploads a medical report via the drag-and-drop interface. The FastAPI backend validates the file, assigns a UUID, and stores it securely before triggering the processing pipeline as a background task.

Stage 2 — Text Extraction and OCR PyMuPDF handles native PDF text extraction. Scanned documents and images are routed to Tesseract OCR with OpenCV preprocessing (grayscale, thresholding, denoising). If a PDF returns insufficient text, OCR fallback is triggered automatically.

Stage 3 — Text Cleaning and Chunking Extracted text is cleaned of OCR artifacts and normalized. The clean text is split into 512-character chunks with 64-character overlap using LangChain's `RecursiveCharacterTextSplitter`, ensuring no clinical sentence is broken across chunk boundaries.

Stage 4 — Embedding and Vector Storage Each chunk is embedded using `all-MiniLM-L6-v2` and stored in a ChromaDB collection namespaced to the report, with cosine similarity configured as the retrieval metric.

Stage 5 — Query and Semantic Retrieval The user's question is embedded using the same model. ChromaDB returns the top five most semantically relevant chunks, ranked by similarity score.

Stage 6 — LLM Response Generation Retrieved chunks, conversation history, and a medical system prompt are passed into GPT-4o via a LangChain LCEL chain. The response streams token-by-token back to the frontend via Server-Sent Events.

Stage 7 — Response Delivery and Citation The answer is displayed with collapsible source citation chips. The full exchange is persisted to the session database and logged in the audit trail.



Figure 6.7 Complete System Workflow Pipeline

6.8 Conclusion

This chapter provided a complete visual and technical walkthrough of all major MedRAG modules through live application screenshots.

- The **Upload Module** confirmed reliable document ingestion across PDF, scanned, and image formats with real-time OCR feedback.
- The **Summary Overview** demonstrated the LLM's ability to synthesize complex lab data into plain, patient-friendly narratives.
- The **Biomarkers Panel** validated the abnormality detection engine with color-coded clinical classifications and visual bar charts.
- The **PDF Export** confirmed the system produces shareable, professionally formatted clinical summaries for real-world consultations.
- The **JSON Response** documentation verified clean, consistent API data contracts across all backend endpoints.
- The **RAG Assistant** proved end-to-end conversational accuracy — with semantic retrieval, streamed responses, and source-cited answers grounded strictly in the uploaded report.

BIBLIOGRAPHY

- [1] LearnersByte Platform, *Bharat Unnati AI Fellowship Training Modules*, Available: <https://learnersbyte.com>, Accessed: April 2026.
- [2] W3School, Python Programming Tutorial. Available at: W3School Python Tutorial (Accessed: April 2026).
- [3] OpenAI, *ChatGPT Documentation*, Available: <https://chat.openai.com>, Accessed: April 2026.
- [4] Notion Labs Inc., *Notion AI User Guide*, Available: <https://www.notion.so/product/ai>, Accessed: April 2026.
- [5] Google, *NotebookLM Documentation*, Available: <https://notebooklm.google.com>, Accessed: April 2026.
- [6] Google Cloud, *Introduction to Generative AI*, Available: <https://cloud.google.com/ai>, Accessed: April 2026.
- [7] Microsoft, *Introduction to Artificial Intelligence Concepts*, Available: <https://learn.microsoft.com>, Accessed: April 2026.
- [8] Visvesvaraya Technological University (VTU), *Internship Guidelines for Engineering Students*, Belagavi, Karnataka, 2024.